

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

How to use this template

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions, everything in green. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

Element C: Presentation and Justification of Solution Design

Requirements

Design Requirement #1 (HIGH Priority)

Watering system must be made out of mold-resistant material

We don't want mold to be growing in the watering system since it will negatively impact human and plant health.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

We got no feedback from our primary stakeholders

Design Requirement #2 (HIGH Priority)

Watering system must be easy to maintain.

It should be easy to maintain for people to not waste their time trying to maintain a hard system.

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No feedback from stakeholders

Design Requirement #3 (HIGH Priority)

The watering system must have an automated scheduling feature for consistent watering.

Research indicates that inconsistent watering leads to plant stress and inefficient water use. An automated schedule ensures plants receive water at optimal times, reducing water waste and improving plant health.

No feedback from stakeholders

Design Requirement #4 (HIGH Priority)

The watering system must be compatible with different plant types by allowing adjustable water flow settings.

Different plants have varying water requirements. Adjustable flow settings ensure that the system can be used for a diverse range of plants, enhancing its versatility. Research on plant hydration needs supports this design feature.

No feedback from stakeholders

Design Requirement #5 (HIGH Priority)

The system must be energy-efficient

Energy-efficient systems reduce costs and environmental impact. Studies show that low-energy irrigation systems are more sustainable and reduce dependency on external power sources.

No feedback from stakeholders

Design Requirement #6 (LOW Priority)

The system should have a mobile app for remote monitoring and control.

While not essential, a mobile app would increase user convenience and accessibility. Research suggests that remote-controlled watering systems lead to better user engagement and system efficiency. This can also be used to monitor the greenhouse to see



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if there are any outside factors that can influence anything such as a rat getting into the greenhouse.

Design Requirement #7 (LOW Priority)

The system should be designed with an aesthetically pleasing exterior that blends well with different garden environments.

While functionality is the priority, aesthetics impact user satisfaction and adoption rates. Research on user preferences indicate s that visually appealing designs are more likely to be integrated into home and public spaces.

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or overly general	are not present or are too vague
C3. Quantity of high-priority design requirements.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None

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_____ of the design requirements listed are identified as high priority.						
C4. Objective & measurable. Design requirements _____.	score = 5 are consistently objective and measurable	score = 4 are nearly always objective and measurable	score = 3 are generally objective and measurable	score = 2 may be only sometimes objective and/or measurable	score = 1 are insufficiently measurable	score = 0 are not measurable
C5. Leading to solution. Design requirements _____.	score = 5 would be highly likely to lead to a tangible and viable problem solution	score = 4 would be likely to lead to a tangible and viable problem solution	score = 3 have a potential to lead to a tangible and viable problem solution	score = 2 have some potential to lead to a tangible and viable problem solution	score = 1 have little potential to lead to a tangible and viable problem solution	score = 0 cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 no primary stakeholder groups.